

Complex numbers

$$z = a + bi$$

Real Imaginary

Complex conjugate

$z = a + bi$ change the sign of Imaginary part

$$z^* = a - bi$$

eg: $3 + 4i \rightarrow z^* = 3 - 4i$

Magnitude

Distance from Zero to complex number

$$|z| = \sqrt{a^2 + b^2}$$

Relation

$$|z|^2 = z z^*$$

eg. $(3 + 4i)(3 - 4i) = 3^2 - 4i^2 = 3^2 + 4 = 25$

$$25 = \underline{\underline{|z|^2}}$$

$|\psi\rangle \Rightarrow$ Quantum State

$|0\rangle \rightarrow$ ket 0

$|1\rangle \rightarrow$ ket 1

eg: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$\alpha, \beta \Rightarrow$ can be complex number

Probability of measuring qubit are

$$P(0) = |\alpha|^2$$

$$P(1) = |\beta|^2$$

Amplitude $\neq$ Probability

Probability = $|\text{amplitude}|^2$

VECTOR

Ordered Collection of number

eg: $\begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightarrow$ components

vector written horizontally - Row vector

vertically - column vector

Row vector $\rightarrow$ related to Bra Notation

column vector $\rightarrow$ related to ket Notation

Transpose

Transpose changes Row $\leftrightarrow$ Column

eg: $(a, b)^T \rightarrow \begin{pmatrix} a \\ b \end{pmatrix}$

Q

$$\begin{pmatrix} a \\ b \end{pmatrix}^T = (a, b)$$

• Vector can contain complex number

eg: $\begin{pmatrix} a + bi \\ c + di \end{pmatrix}$

Connection to Qubit

A Single Qubit is represented by 2D vector

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

General Qubit state is

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

or

For a valid normalised Qubit

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$(|\alpha|^2 + |\beta|^2) = 1$$

Dual notation

Ket - $|\psi\rangle \rightarrow$ Column vector

Bra - $\langle\psi| \rightarrow$ Row vector

Step 1: Transpose

Turn Column $\rightarrow$ Row

$$\begin{array}{c} |\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \\ \downarrow \\ \begin{bmatrix} a & b \end{bmatrix} \end{array} \quad \left. \vphantom{\begin{array}{c} |\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \\ \downarrow \\ \begin{bmatrix} a & b \end{bmatrix} \right\} \right\} \text{Transpose}$$

Step 2: Conjugated

eg: $|\psi\rangle = \begin{bmatrix} 2 + 3i \\ 4 - 5i \end{bmatrix}$

Turn to row matrix or Bra Notation

1. Transpose

$$\begin{bmatrix} 2+3i & 4-5i \end{bmatrix} \quad \langle\psi| \times |\psi\rangle$$

2. conjugate

$$\langle\psi| = \begin{bmatrix} 2-3i & 4+5i \end{bmatrix}$$

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \xrightarrow{\text{bra}} \begin{bmatrix} a^* & b^* \end{bmatrix}$$

When bra & ket multiplies

$$\begin{bmatrix} a \\ b \end{bmatrix} [a \ b]$$

For a Properly normalized Qubit State

$$\langle \psi | \psi \rangle = 1$$

Inner Product

BRA $\times$ KET

$$\langle a | b \rangle$$

$$\langle a | \times | b \rangle$$

$$\langle a | = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$| b \rangle = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

then BRA $\times$ KET

$$\langle a | = [1 \ 2]$$

$$| b \rangle = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\langle a | \times | b \rangle = [1 \ 2] \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$1 \times 3 + 2 \times 4$$

$$= 11$$

$$\langle a | b \rangle = 11$$

$$kg: \langle 0 | 1 \rangle$$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\langle 0 | = [1 \ 0]$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\langle 0 | 1 \rangle = [1 \ 0] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

now

$$\langle 1 | 0 \rangle$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0$$

now

$$\langle 0 | 0 \rangle = [1 \ 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$$

now

$$\langle 1 | 1 \rangle = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1$$

So according to all the Kr inner.

Same state $\times$ same state = 1

Different state = 0 $\rightarrow$ hence Orthogonal

To check two vectors are orthogonal

we take Innerproduct = 0 i.e. $\text{BRA} \times \text{KET} = 0$

Outer product

KET $\times$ BRA $\rightarrow$ matrix

Inner prod $\rightarrow$ number

Outer prod $\rightarrow$ matrix

an Important Example

Let's calculate

$$|0\rangle\langle 0|$$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\langle 0| = [1 \ 0]$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} [1 \ 0]$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$\rightarrow$ Related to
Projecting a Qubit into state $|0\rangle$

Probability Amplitude

$|\psi\rangle = a|0\rangle + b|1\rangle \rightarrow$ P.A. associated with
Prob amplitude associated with $|0\rangle$

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \begin{matrix} \rightarrow \text{for } 0 \\ \text{for } 1 \end{matrix}$$

$$\text{Probability} = |\text{amplitude}|^2$$

Born Rule

$$P(0) = |a|^2$$

$$P(1) = |b|^2$$

Qubit Superposition

Superposition is a Quantum state that is linear combination of two basis states

QUANTUM GATE

→ Basically a matrix

- Quantum computing equivalent of a logic gate

→ in classic computer, gate manipulate bit
in Quantum Computer, Quantum gate manipulates qubit

Gates

Pauli's gate → Wolfgang Pauli, X, Y, Z

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Pauli's X = NOT gate

eg: Suppose Our qubit is

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$X |0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{pmatrix} 0 \times 1 + 1 \times 0 \\ 1 \times 1 + 0 \times 0 \end{pmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

So when we apply X to $|0\rangle$ we get $|1\rangle$

Pauli's $-Y$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Apply Y to $|0\rangle$

$$Y|0\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} =$$

$$= (0 \times 1) + (-i \times 0) = 0$$

$$= (i \times 1) + (0 \times 0) = i \quad \cdot \begin{bmatrix} 0 \\ i \end{bmatrix}$$

as we know $|1\rangle = \begin{bmatrix} 0 \\ i \end{bmatrix}$

$$\therefore Y|0\rangle = i|1\rangle$$

So Y makes $|0\rangle$ to $|1\rangle$ But add factor i

Phase change

Apply Y to $|1\rangle$

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 \\ i \end{bmatrix} = \begin{matrix} (0 \times 0) + (-i \times i) = -i \\ (i \times 0) + (0 \times i) = 0 \end{matrix} \begin{bmatrix} -i \\ 0 \end{bmatrix}$$

$$\text{So } |1\rangle = -i|0\rangle$$

Pauli's Z

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Apply Z on $|0\rangle$

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} (1 \times 1) + (0 \times 0) &= 1 \\ (0 \times 1) + (-1 \times 0) &= 0 \end{aligned} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

No change

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned} (1 \times 0) + (0 \times 1) &= 0 \\ (0 \times 0) + (-1 \times 1) &= -1 \end{aligned} \quad \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\underline{\underline{Z|1\rangle = -|1\rangle}}$$

HADAMARD H

creates superposition.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Apply H to $|0\rangle$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= (1 \times 1) + (1 \times 0) = 1$$

$$(1 \times 1) + (-1 \times 0) = 1$$

$$= \frac{1}{\sqrt{2}} \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$